END OF SECTION

Name: Solutions

Teacher: ____________________

Class: ____________________

Baldivis
Secondary College

MATHEMATICS – PATHWAY 1
YEAR 10 EXAM
SEMESTER ONE 2020

Section One: Resource-Free

TIME ALLOWED FOR THIS PAPER

Reading time before commencing **5 minutes.**
Working time for paper **40 minutes.**

MATERIAL(S) REQUIRED / RECOMMENDED FOR THIS PAPER

TO BE PROVIDED BY THE SUPERVISOR: This Question / Answer Booklet Formula Sheet

TO BE PROVIDED BY THE CANDIDATE: Standard items: pens, pencils, pencil sharpener, eraser, correction fluid, ruler, highlighters. Special items: nil **Important note to candidates:** No other items may be used in this section of the examination. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Instructions to candidates

- Candidates are to remain seated during the exam and are not allowed to talk to other candidates.
- Mobile phones are not to be used at all during the exam.
- Students who fail to follow examination rules will be removed and dealt with in accordance to the schools policy.

Write your answers in the spaces provided in this Question/Answer Booklet. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

- Planning: If you use spare pages for planning, indicate this clearly at the top of the page
- Continuing an answer: If you need the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

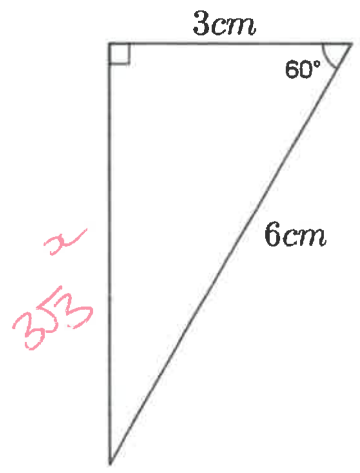
Show all your working clearly. Your working should be sufficient to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat an answer to any question, ensure that you cancel the answer you do not wish to have marked.

Use a blue or black pen only, except with diagrams where you should use pencil.

Question	*Out of*	*Mark*
1	6	
2	10	
3	10	
4	9	
5	5	
TOTAL	***40***	
	Mark	
	Percentage	

Question 5 **(5 marks)**

a) What is the value of cos 60° for the triangle below? (1 mark)



$\cos 60° = \frac{1}{2}$

$x = \sqrt{6^2 - 3^2}$

$= \sqrt{36 - 9}$

$= \sqrt{27}$

$= 3\sqrt{3}$

$\cancel{\cos 60° = \frac{3\sqrt{3}}{6}}$

$\cancel{= \frac{\sqrt{3}}{2}}$

b) A similar triangle has an hypotenuse of 20cm. What are the lengths of the other two sides? (4 marks)

10cm.

$\frac{20}{6} = \frac{10}{3}$

$3\sqrt{3} \times \frac{10}{3} = 10\sqrt{3}$

$\sqrt{400 - 100}$

$= \sqrt{300}$

$= 10\sqrt{3}$

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Question 4 **(9 marks)**

a) Which of the following are equal to $5\sqrt{3}$ **(5 marks)**

$2\sqrt{3}+3\sqrt{3}$ ✓	$\sqrt{25}\times\sqrt{9}$ ✗	$\sqrt{75}$ ✓
$3\sqrt{3}+\sqrt{12}$ ✓	$\sqrt{15}$ ✗	$\sqrt{5}\times\sqrt{15}$ ✓

b) Simplify the following

i) $\sqrt{5}\times\sqrt{10}$ (2 marks)

$\sqrt{50} = 5\sqrt{2}$

ii) $3\sqrt{2}+5\sqrt{8}$ (2 marks)

$3\sqrt{2} + 10\sqrt{2} = 13\sqrt{2}$

Question 1 **(6 marks)**

Factorise the following

a) $6x-20x^2$ (1 mark)

$2x(3x-10x)$

b) x^2-100 (1 mark)

$(x-10)(x+10)$.

c) $x^2-7x-\cancel{100}$ 30 (2 marks)

$(x-10)(x+3)$.

d) $3x^2-7x-20$ (2 marks)

$3x^2-12x+5x-20$

$= 3x(x-4)+5(x-4)$

$= (3x+5)(x-4)$.

$\times 60$

1x60
2x30
3x20
4x15
5x12

Question 2 **(10 marks)**

a) Evaluate

i) $25^{\frac{1}{2}}$ $= 5$ (1 mark)

ii) $27^{\frac{2}{3}}$ $= 9$ (2 mark)

b) Simplify each expression, write answers with positive indices.

i) $4x^3y \times (3xy)^2$ (2 marks)

$36x^5y^3$.

ii) $\dfrac{5a^{-2}b^5}{2c^{-1}}$

$\dfrac{5b^5c}{2a^2}$

(2 marks)

iii) $\left(\dfrac{3p^2q^4}{2p^{-3}q^{-2}}\right)^{-2}$ $\dfrac{4}{9p^{10}q^{12}}$

(3 marks)

$\left(\dfrac{2p^{-3}q^{-2}}{3p^2q^4}\right)^2$

$\dfrac{2p^{-6}q^{-4}}{3p^4q^8}$.

Question 3 **(10 marks)**

Simplify

a) $\dfrac{3t}{4} - \dfrac{t}{5}$ $= \dfrac{15t}{20} - \dfrac{4t}{20} = \dfrac{11t}{20}$

(2 marks)

b) $\dfrac{6h}{5} \times \dfrac{h}{3}$ $\dfrac{2h^2}{5}$.

(2 marks)

c) $\dfrac{5k+2}{3} + \dfrac{k}{2}$ $\dfrac{10k+4+3k}{6} = \dfrac{13k+4}{6}$

(3 marks)

a) $\dfrac{3x+2}{3} \div \dfrac{x+4}{2}$

(3 marks)

$\dfrac{6x+4}{3x+12}$